

Serie 10

LINEAR MAPS

1. Choice of basis. Show that if V is a one-dimensional vector space over a field K and $T \in \text{Hom}_K(V, V)$, then there exists $\lambda \in K$ such that for all $v \in V : Tv = \lambda v$. Explain then why an isomorphism $V \rightarrow K$ depends on a choice of basis, while one from $\text{Hom}_K(V, V)$ to K doesn't.

2. Linear map on a subspace. Let V, W be vector spaces over a field K . Suppose that $U \subsetneq V$ is a linear subspace and let S be a non-trivial element of $\text{Hom}_K(U, W)$ (i.e. we assume that S does not map everything to 0). Define $T : V \rightarrow W$ by

$$Tv = \begin{cases} Sv, & \text{if } v \in U \\ 0, & \text{if } v \in V \setminus U \end{cases}$$

Is T a linear map?

3. Linear map over a subspace. Let U, V, W be vector spaces over a field K and let $T : V \rightarrow W$ and $S : W \rightarrow U$ be linear maps.

(a) Prove that

$$\text{rank}(S \circ T) \leq \min(\text{rank}(S), \text{rank}(T)).$$

(b) Show that $\text{rank}(S \circ T) = \text{rank}(S)$ whenever T is surjective.

(c) Show that $\text{rank}(S \circ T) = \text{rank}(T)$ whenever S is injective.

4. Projection. Let V be a vector space. An endomorphism $P : V \rightarrow V$ satisfying $P^2 := P \circ P = P$ is called idempotent or a projection. Show:

(a) For every projection P , its image $\text{Im}(P)$ is a linear complement of $\text{Ker}(P)$ in V .

(b) For any subvectorspaces $W_1, W_2 \subset V$, such that W_1 is a complement of W_2 in V , there exists a unique projection $P : V \rightarrow V$ with

$$\text{Ker}(P) = W_1 \quad \text{und} \quad \text{Im}(P) = W_2.$$

5. Preimage of a linear map. Let $f : V \rightarrow W$ be a linear map of K -vector spaces. Show:

- (a) For every subvectorspace $W' \subset W$ the preimage

$$f^{-1}(W') := \{v \in V \mid f(v) \in W'\}$$

is a subvectorspace of V .

- (b) We have

$$\dim f^{-1}(W') = \dim \operatorname{Ker}(f) + \dim (\operatorname{Im}(f) \cap W').$$

6. Isomorphisms of vector spaces. Let V, W be vector spaces over a field K and let $T : V \rightarrow W$ be an isomorphism of vector spaces. Show that:

- (a) T maps linearly independent sets to linearly independent sets;
- (b) T maps spanning sets of V to spanning sets of W ;
- (c) T maps bases to bases.

7. Dimension and basis. Let V, W be vector spaces over $\mathbb{Q}$. We say that a map $f : V \rightarrow W$ is additive, if

$$\forall x \in V \forall y \in V : f(x + y) = f(x) + f(y).$$

Show that

$$\operatorname{Hom}_{\mathbb{Q}}(V, W) = \{f : V \rightarrow W \mid f \text{ is additive}\}.$$